

India's Clean Tech Vanguard: Drumil Joshi on AI Powering the Global Energy Transition

In a business landscape where energy security and sustainability have become strategic priorities, one Indian technologist stands at the forefront of innovation. Drumil Joshi is a globally respected **expert in artificial intelligence applied to renewable energy operations. His work combines technical rigor, real-world deployment, and measurable economic value.** With a portfolio of renewable assets under his supervision valued at \$450 million, his insights are reshaping how clean power is **forecasted and delivered. This profile highlights why his contributions have attracted international recognition and why his story matters to India's economic future.**



Drumil Joshi

Expert in Artificial Intelligence

Drumil Joshi in an interview with Salil Parekh

How did your journey with AI and renewable energy begin?

"Growing up in Mumbai and Gujarat, I saw firsthand how volatile power supply affects daily life. When I pursued data science, I realized that advanced analytics could make renewable energy not only abundant but reliable. Traditional forecasting and maintenance are reactive. My goal was to make them predictive and strategic."

Joshi's approach uniquely integrates machine learning, meteorology, and operational engineering to solve complex energy challenges. His models achieve prediction errors as low as 1 to 2 percent, a benchmark that industry analysts describe as setting new standards for accuracy in renewable forecasting.

Your innovations have been featured internationally. What sets them apart?

"It is not just about algorithms. It is about impact. For example, our predictive maintenance tools and outage-scheduling systems reduce avoidable energy losses and extend asset life. We created an ensemble intelligence platform that prioritizes issues before they escalate and integrates smoothly into real-time operations."

His work has been spotlighted in global media and technical communities, demonstrating the economic and operational benefits of AI integration in energy systems. His forecasting work has drawn academic citations and extensive reads on research platforms, reflecting widespread professional engagement.

What does your role at Southern Power involve?

"At Southern Power, I monitor analytics for 34 utility-scale solar plants, 15 wind farms, and 2 flagship battery systems.

These assets operate across diverse climates and market conditions. My team and I apply real-time diagnostics to improve performance and reduce downtime."

Industry peers note that Joshi's platform recovers significant energy otherwise lost to inefficient scheduling or unplanned outages, contributing measurable financial value to operators.

How do you see the future of AI in energy markets evolving?

"AI must be practical, transparent, and scalable. Future systems will not only forecast but autonomously guide decisions that balance generation, storage, and distribution. India's ambitious renewable targets require such intelligent systems to ensure that every watt counts."

As India scales its clean energy capacities and explores next-generation markets like green hydrogen and distributed grids, leaders like Drumil Joshi exemplify how Indian innovators are shaping global sustainability outcomes with technology and strategy.

